Module 08: Planning in Embodied Cognition — Somatic Anticipation

Learning Objectives

1. Define **somatic anticipation** as the embodied agent's capacity to plan future actions through bodily simulation rather than abstract deliberation.
2. Analyze how **motor imagery, prospective body states, and anticipatory postural adjustments** implement planning in Active Inference.
3. Apply the embodied planning framework to understand **expertise in time-critical domains** (sports, surgery, music, emergency response).

Introduction

Planning, in the Cartesian view, is a purely cognitive process — the mind contemplates possible futures and selects the best one. Embodied cognition reveals that planning is fundamentally a **somatic process**: the body simulates future actions, generates predicted sensory consequences, and selects policies based on the *felt quality* of those predictions.

A tennis player planning a return doesn't run a mental simulation in abstract coordinates — their body *pre-positions*, shifting weight and adjusting grip before the ball arrives. A pianist planning a difficult passage doesn't merely compute the notes — their fingers pre-rehearse the movement pattern in motor imagery. Planning is prediction running through the body.

Key Concepts

1. Motor Imagery as Embodied Simulation

Motor imagery — the mental rehearsal of movement without actual execution — engages the same neural substrates as actual movement (primary motor cortex, supplementary motor area, cerebellum):

- Motor imagery generates predicted proprioceptive trajectories without unleashing the descending commands that would activate muscles
- The quality of motor imagery correlates with motor expertise — experts generate more vivid, precise motor simulations
- Active Inference interpretation: Motor imagery is the evaluation of **Expected Free Energy** for candidate policies — the agent simulates the proprioceptive consequences of each possible action sequence and selects the one with lowest EFE

2. Anticipatory Postural Adjustments (APAs)

Before any voluntary movement, the body makes **anticipatory postural adjustments** — pre-emptive shifts in muscle activation that prepare for the perturbation the planned movement will create:

- Before lifting an arm, the legs and trunk pre-stabilize to counteract the shift in center of mass
- APAs occur 50-100ms before the voluntary movement — too fast for conscious planning
- Active Inference interpretation: APAs are the proprioceptive predictions of the postural generative model — the body predicts the consequences of the planned action and pre-emptively minimizes the postural prediction error

3. Temporal Horizon and Embodied Expertise

Expert embodied planners have longer **temporal horizons** — they simulate further into the future:

- A novice chess player (embodied metaphor notwithstanding) plans 1-2 moves ahead
- A novice climber plans 1-2 handholds ahead
- An expert climber plans entire sequences, feeling the whole route as a kinesthetic gestalt
- An expert surgeon plans the entire procedure, anticipating complications before they arise

In Active Inference, temporal horizon depth is determined by the **hierarchical depth** of the generative model — deeper models encompass longer timescales.

4. Somatic Markers and Prospective Evaluation

When evaluating future actions, the embodied agent doesn't merely compute expected utility — it generates **predicted somatic states** (Damasio's somatic markers):

- "Imagining presenting at the conference makes my stomach tighten" → the body simulates the interoceptive consequences of the planned action, providing a gut-level evaluation
- These somatic predictions are *genuine* predictions — they activate the same autonomic circuits that the real situation would activate
- The felt quality of the predicted state (comfortable, anxious, excited) informs policy selection directly, without requiring propositional evaluation

Applications

- **Athletic visualization**: Elite athletes use motor imagery to pre-rehearse competition performances — activating the embodied planning system to refine proprioceptive predictions before the event. Systematic motor imagery practice improves actual performance across sports from gymnastics to golf.
- **Emergency response planning**: Experienced emergency responders plan by running somatic simulations of scenarios — "If I enter through this door, I'll face heat from the

left; if through that window, I'll need to drop two meters." The planning is kinesthetic, not abstract, and the quality of the plan depends on the depth of the embodied generative model.

Conclusion

Planning in embodied cognition is somatic anticipation — the body simulates future actions, generates predicted sensory consequences, and evaluates policies through felt quality. Motor imagery, anticipatory adjustments, temporal horizon depth, and somatic markers all implement Active Inference planning through bodily channels. This completes the Intuitive Knowing unit — the next unit, Moving Through World, examines how embodied agents navigate, interact with, and transform their physical environments.